

Timothy Medewase

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EDUCATION

University of Montevallo

Bachelor of Science in Computer Science, Minor in Mathematics — GPA: 3.8/4.0

Montevallo, AL

August 2021 - May 2025

EXPERIENCE

Software Engineer (AI/ML)

May 2025 – Present

Figbox Technologies, Inc

Upper Marlboro, Maryland

- Developed ML-powered backend services for the Figbox platform supporting **20,000+** active users through data pipelines, inference workflows, and containerized deployments using **Django**, **Docker**, and **DigitalOcean**
- Built a natural-language search feature for user profiles by combining **spaCy**-based text processing with **OpenAI API** for query understanding and ranking, integrated into a full-stack platform using **React**, **TypeScript**, and **Node.js**

Software Development Intern

May 2023 – Jul. 2023

Solera, Inc

Westlake, TX

- Developed a full-stack analytics dashboard using **React** and **d3.js**, building reusable data visualization components for internal reporting and analytics workflows
- Engineered backend data workflows with **Microsoft SQL Server**, optimizing stored procedures and query execution to reduce query latency by **50%**; integrated **RESTful API** endpoints via **Axios** and contributed to **CI/CD** workflows using **Bitbucket Pipelines**

PROJECTS

Deep Research Agent | *Next.js, FastAPI, Python, OpenAI, Chroma, Exa, LangChain* | Link

Apr. 2026

- Built a full-stack research agent that decomposes natural-language queries into subquestions, runs iterative tool-calling loops, and synthesizes cited final responses using **OpenAI**, supported by a constraint-based orchestration layer enforcing token, cost, and time budgets to ensure reliable execution
- Designed a streaming SSE event system (**thinking**, **tool_call**, **memory_hit**, **answer**) with a **Chroma**-backed vector memory store, enabling real-time visibility into agent reasoning and persistent context reuse across multi-step workflows

OpenBolts | *FastAPI, Next.js, React, Supabase, Docker* | Link

Jan. 2026

- Built an OEM knowledge layer for automotive repair using **FastAPI**, **Next.js**, **React**, and **Supabase**, implementing a hybrid **RAG** pipeline that ingests manufacturer manuals and delivers page-level citations with sub-**500ms** average query response time
- Implemented backend retrieval pipelines with hybrid scoring (semantic + structural + keyword + grep), bounded refinement, context rewriting, and **Exa** web fallback to improve retrieval relevance and reduce out-of-scope responses compared to baseline keyword search
- Developed full-stack operational tooling including real-time monitoring dashboards and per-phase ingestion tracking to increase system visibility, speed up issue detection, and support reliable deployment at scale

SCOUR | *Swift, RevenueCat, Supabase, FastAPI* | Link

Oct. 2025

- Shipped an **iOS** recruiting app using **Swift**, **RevenueCat**, and **Supabase** for athlete-coach matching, enabling athletes to browse and filter **15,000+** college programs by sport, division, and location
- Integrated an AI-assisted outreach and CRM backend with **FastAPI** and **Supabase** to manage subscriptions, track coach contacts, and generate outreach emails while syncing hundreds of athlete and coach records

Transformer Implementation | *Python, PyTorch, Deep Learning* | Link

May 2025

- Implemented the full Transformer architecture from the ground up in **PyTorch** with **6-layer** encoder-decoder stacks, **8 attention heads**, and **512-dimensional** embeddings following Vaswani et al.
- Developed modular components including multi-head attention, residual connections, and sinusoidal positional embeddings, enabling configurable depth, width, and head count via a base/big setup
- Optimized attention computations and tensor reshaping to improve training efficiency and support autoregressive decoding

TECHNICAL SKILLS

Languages: Java, TypeScript/JavaScript, SQL, Python, C++, Swift, HTML/CSS

Frameworks & Libraries: React.js, Redux, Next.js, Node.js, Express.js, FastAPI, Django, Flask, Spring Boot, Pandas, NumPy, TensorFlow, PyTorch, Scikit-learn, LangChain, TailwindCSS, Material-UI

Tools, Databases & Deployment: Git, GitHub, Docker, Kubernetes, AWS, Vercel, DigitalOcean, CI/CD (Bitbucket Pipelines, GitHub Actions), PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Supabase, SQLAlchemy, REST APIs, GraphQL, FAISS, OpenAI API, Exa API